Cereal 1 — Guide

Oct 26, 2026 · Anas Alsirafy

The cereal track detects foliar diseases on field photos of maize, where one photo can show several diseases at once. The best model so far is a single ConvNeXt-Base with per-label thresholds (macro-F1 0.468); a five-specialist redesign scored lower (0.283) and is unresolved. This set is for a new team member with basic AI knowledge.

Where this sits

This handover has three parts. This set covers maize. The cassava classifier, where the training pipeline was first developed, has its own eight docs starting at *Cassava 1 — Guide*. The five-dataset comparison behind Paper 2 is one doc: *Cross-Dataset Study — Five Datasets (Paper 2)*. Rice is a cereal but was only used in that comparison, so it lives there.

Status

Piece	State (Oct 2026)
Monolithic multi-label model (`maize_multilabel.ipynb`)	Done. Macro-F1 0.468, micro-F1 0.543, Hamming loss 0.204 (5-fold, tuned thresholds).
Specialist pipeline (`maize_specialist.ipynb`)	Run on all folds. Macro-F1 0.283 (tuned). Worse than monolithic. A bug in how it saves models is a likely partial cause.
Literature	No prior deep-learning benchmark found on this 8-label field dataset.
Paper	Not started.

How to read this set

1. *Cereal 2 — Environment Setup* — hardware, packages, the checkpoint pattern for long runs.
2. *Cereal 3 — Data* — the Wiesner-Hanks 2018 dataset, its labels and their counts.
3. *Cereal 4 — Configs* — settings for both pipelines.
4. *Cereal 5 — Experiments* — what was run and why.
5. *Cereal 6 — Results and Findings* — the numbers and what they mean.
6. *Cereal 7 — Known Issues and Gotchas* — bugs and traps. Read before touching code.
7. *Cereal 8 — Future Work* — what to do next.

Glossary

Term	Meaning here
Multi-label	An image can have several labels at once. The model has one sigmoid output per label, each answering yes/no independently.
Weighted BCE	Binary cross-entropy where positives of rare labels count more: pos_weight = (N − positives) / positives, capped at 50.
Threshold tuning	Choosing, per label, the probability above which we say "yes" (instead of 0.5).
OOF predictions	Each image's prediction from the fold where it was held out.

Term	Meaning here
Macro-F1 / micro-F1	Macro averages F1 over labels equally; micro pools all decisions, so common labels dominate.
Hamming loss	Fraction of all (image, label) decisions that are wrong. Lower is better.
Monolithic	One network predicting all 8 labels.
Specialist	A smaller model trained on one sub-problem (e.g. healthy vs gray leaf spot); outputs are combined.
Three-phase training	Frozen backbone → top 30 layers unfrozen → whole network unfrozen.
MultilabelStratifiedKFold	A fold splitter that keeps every label's share roughly equal across folds.

People and files

Who / what	Detail
Anas Alsirafy	Built this work; leaving AUC (anas.samy@aucegypt.edu)
Prof. Maki K. Habib	Supervisor
Successor	Not yet chosen
Notebooks	`maize_multilabel.ipynb`, `maize_specialist.ipynb`; generated by `build_maize_nb.py` and `build_specialist_nb.py` (the builder scripts are the source of truth)
Labels	`Database.csv`
Location	Google Drive folder, link to be added

Open questions for Anas

- [ ] Add the Drive link.
- [ ] Confirm where saved maize models, `specialist_checkpoint.json` and the OOF `.npy` files are.
- [ ] Confirm the GPU used for maize runs (RTX 5090 or Kaggle).